

Does Smartphone Use Before Sleep Affect Sleep Duration?

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Abstract—This study examines whether smartphone use before sleep affects sleep duration among college students. Survey data were collected from 28 college students regarding their smartphone usage within 30 minutes before sleep and their sleep duration. Welch’s two-sample t-test was conducted to compare the mean sleep duration between students who used smartphones before sleep and those who did not. The results showed a significant difference in sleep duration between the two groups ($p = 0.00083 \approx 0.001$). Students who used smartphones before sleep had a lower average sleep duration (5.43 hours) than those who did not use smartphones before sleep (7.21 hours). Therefore, students who used smartphones before sleep tended to have shorter sleep durations than those who did not.

I. INTRODUCTION

Sleep is an important factor affecting students’ health and academic performance. Smartphone use before sleep has become increasingly common among college students, and this study investigates whether it is associated with differences in sleep duration.

1.1 Research Question

The primary research question of this study is:
Is there a difference in sleep duration between students who use smartphones within 30 minutes before sleep and those who do not?

1.2 Study Objective

The objective of this study is to investigate whether smartphone use within 30 minutes before sleep is associated with differences in sleep duration among college students.

II. DATA COLLECTION AND DESCRIPTION

We collected data from 28 college students using a Google Forms survey.

2.1 Data Collection Method

The survey followed these simple steps:

1. Participants were asked whether they used smartphones within 30 minutes before sleep on Sunday night.
2. Participants reported their actual sleep duration on Sunday night.
3. Participants completed a short demographic question.
4. Responses were recorded using Google Forms.

2.2 Dataset Description

A total of 28 valid responses were collected.

- Smartphone users: 14 participants
- Non-smartphone users: 14 participants
- Explanatory Variable (X): Smartphone use within 30 minutes before sleep (Yes / No)
- Response Variable (Y): Sleep duration (hours)
- Additional Variable: Academic year (Freshman, Sophomore, Junior, Senior, Graduate Student)

The analysis focused on the relationship between smartphone use before sleep and sleep duration.

The two groups were compared based on their average sleep duration.

2.3 Data Cleaning

Before analysis, all responses were reviewed for completeness and consistency.

- No incomplete responses were identified.
- No unclear or invalid responses were identified.
- Smartphone use before sleep was coded as a categorical variable (Yes / No).
- Sleep duration was recorded as a numerical variable measured in hours.

After data cleaning, 28 valid responses were retained for analysis.

III. VARIABLES AND STUDY DESIGN

3.1 Variables

1. Explanatory Variable (X): Smartphone use within 30 minutes before sleep (Yes / No).
2. Response Variable (Y): Sleep duration (hours).

3.2 Study Type

This study follows a comparison (difference between groups) design. Participants were divided into two independent groups based on whether they used smartphones within 30 minutes before sleep.

IV. METHOD SELECTION

4.1 Planned Statistical Method

We used Welch’s two-sample t-test.

4.2 Justification

We chose this method because we compared the mean sleep duration between two independent groups (students who used smartphones before sleep and students who did not). The explanatory variable (smartphone use before sleep) is categorical with two groups, while the response variable (sleep duration) is a numerical variable. Therefore, Welch’s two-sample t-test is an appropriate method for this study.

V. ANALYSIS RESULTS

5.1 Descriptive Statistics

Group	n	Mean	SD
Smartphone Users	14	5.429	0.646
Non-Smartphone Users	14	7.214	1.528

Students who used smartphones before sleep had a lower average sleep duration (5.43 hours) than students who did not use smartphones before sleep (7.21 hours).

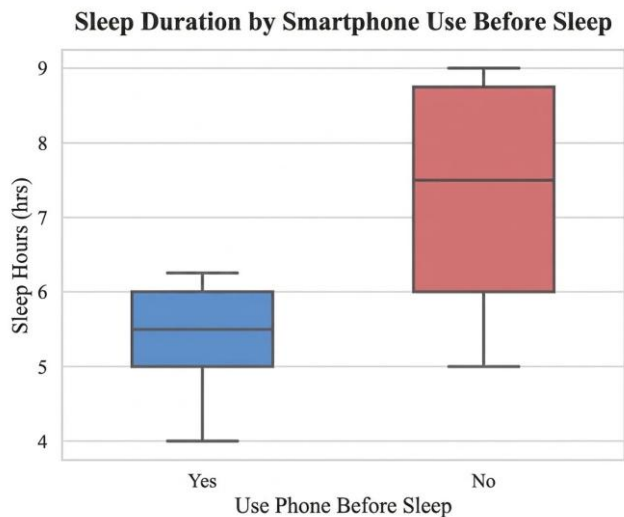
5.2 Welch's Two-Sample T-Test

Item	Value
t-statistic	-4.027
Degrees of Freedom	17.505
p-value	0.001

Because the p-value (0.001) is less than 0.05, the null hypothesis was **rejected**.

5.3 Visualization

Fig. 1. Boxplot of Sleep Duration by Smartphone Use



VI. CONCLUSION AND LIMITATIONS

5.1 Conclusion

Because the p-value (0.001) is less than 0.05, we reject the null hypothesis. There is a significant difference in sleep duration between students who use smartphones before sleep and those who do not. Students who used smartphones before sleep had a lower average sleep duration (5.43 hours) than students who did not use smartphones before sleep (7.21 hours). Therefore, smartphone use before sleep may be associated with shorter sleep duration among college students.

5.2 Limitations

1. Sample Size: Only 28 participants were included in this study. A larger sample size may provide more reliable results.
2. Self-Reported Data: Sleep duration was self-reported by participants and may not perfectly reflect actual sleep time.
3. Other Factors: Factors such as academic workload, stress, caffeine consumption, and individual sleeping habits were not controlled in this study.

EXTRA: APPENDICES

APPENDIX A: DATASET SAMPLE

Smartphone Use	Sleep Duration	Academic Year
Yes	5	Junior
Yes	6	Sophomore
Yes	4	Sophomore
No	8	Freshman
No	9	Senior

APPENDIX B: PROJECT POSTER DRAFT

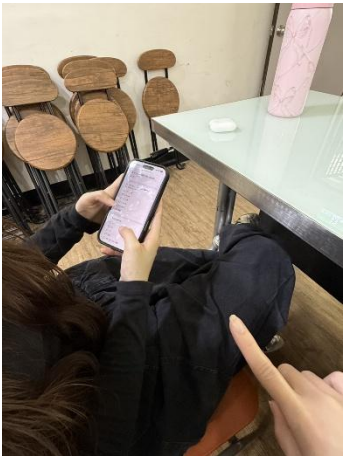
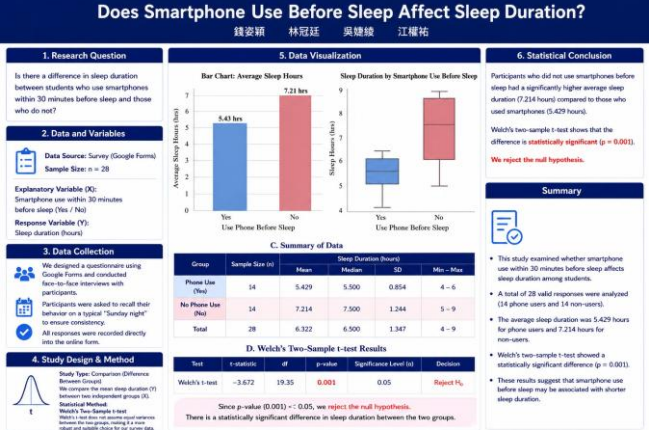


Fig. C3. Data collection from a sophomore participant.



Fig. C4. Data collection from a junior participant.

APPENDIX C: EVIDENCE PHOTOS



Fig. C1. Data collection from a sophomore participant.



Fig. C2. Data collection from a senior participant.



Fig. C5. Data collection from a sophomore participant.



Fig. C7. Data collection from a junior participant.



Fig. C8. Data collection from a senior participant.



Fig. C6. Data collection from a sophomore participant.